# CURRICULUM VITAE

Dr Ramachandran Thiruvengadam
M.B.,B.S., DCH, M.D. (Biochemistry), Ph.D. (Maternal and Child Health)
Associate Professor, Maternal and Child Health Programme, BRIC-Translational Health Science and Technology Institute (THSTI), Faridabad, Haryana, India

## 1. Contact and identifiers



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|---|---|
| **Address** | NCR Biotech Science Cluster, 3rd Milestone, Faridabad-Gurugram Expressway, Faridabad 121001, Haryana, India |
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| **Web of Science ID** | AFK-3492-2022 |
| **Google Scholar** | https://scholar.google.com/citations?user=2yFI5XoAAAAJ |
| **Domain expertise** | Maternal and child health; perinatal epidemiology; clinical trials; clinical prediction models, AI-enabled imaging tools and multi-omics for adverse pregnancy outcomes; clinical and translational immunology |



## 2. Profile

Physician-scientist and translational researcher in maternal and child health. I develop and validate predictive and diagnostic tools and interventions that can be deployed at public-health scale to reduce adverse pregnancy outcomes such as preterm birth, fetal growth restriction and the great obstetrical syndromes, through robustly designed cohort studies, randomised trials and secondary analyses of pooled national data. I am an investigator in the Interdisciplinary Group for Advanced Research on Birth Outcomes - DBT India Initiative (GARBH-Ini), where I work with multi-omics and data-science teams to discover biomarkers and build AI-driven clinical decision tools, and I have translated this work into four Indian patent applications (three awarded). During 2020-2023 I coordinated the clinical research component of the DBT Consortium for COVID-19 Research, including the real-world effectiveness evaluation of COVID-19 vaccines that informed national immunisation policy. I currently lead, as Principal Investigator, two ICMR National Health Research Priority (NHRP) projects on hypothyroidism in pregnancy and gestational weight gain, and I am Co-Principal Investigator of the ICMR-funded GRASPP randomised controlled trial and the Gates Foundation-supported MOMI-India consortium.

## 3. Academic qualifications

| Institution | Degree | Year |
|---|---|---|
| Translational Health Science and Technology Institute, Faridabad (affiliated to Jawaharlal Nehru University, New Delhi) | Ph.D. (Life Sciences - Maternal and Child Health) | 2023 |
| All India Institute of Medical Sciences (AIIMS), New Delhi | M.D. (Biochemistry) | 2016 |
| Institute of Child Health and Hospital for Children, Madras Medical College, The Tamil Nadu Dr MGR Medical University | Diploma in Child Health (DCH) | 2009 |
| Madurai Medical College, Madurai, The Tamil Nadu Dr MGR Medical University | M.B.,B.S. | 2005 |

## 4. Specialised training and courses

| Course | Place and date | Conducted by |
|---|---|---|
| EpiCap pedagogy workshop: analytic epidemiology for causal inference, | Bergen, Norway; 16-23 February 2024 | University of Bergen |

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| teaching for learning | | |
| Fundamental concepts in medical statistics with applications to RCTs (co-facilitator) | Kampala, Uganda; 7-13 June 2023 | University of Bergen; Makerere University |
| Advanced epidemiology and causal inference | Bergen, Norway; 25-29 April 2022 | University of Bergen |
| Introduction to modern epidemiology and clinical research methodology | Bergen, Norway; 20-22 April 2022 | University of Bergen |
| Short course series on environmental exposure assessment | PHFI, Gurugram; Oct and Dec 2018 | ISEE, PHFI, CCDC |
| NIH-ICMR-THSTI course on principles and practice of clinical research | Hyderabad; 16-21 April 2018 | NIH, USA and ICMR |
| Data Science specialisation (online) | June-September 2017 | Johns Hopkins Bloomberg School of Public Health |

## 5. Professional experience

| Institution | Position | From | To |
|---|---|---|---|
| BRIC-Translational Health Science and Technology Institute (THSTI), Faridabad | Associate Professor, Maternal and Child Health Programme | 30 June 2025 | Till date |
| Pondicherry Institute of Medical Sciences, Puducherry | Assistant Professor, Department of Biochemistry | May 2022 | June 2025 |
| THSTI, Faridabad | Principal Scientist - I | November 2021 | May 2022 |
| THSTI, Faridabad | Senior Research Scientist | July 2021 | November 2021 |
| THSTI, Faridabad | Scientist D | January 2020 | June 2021 |
| THSTI, Faridabad | Scientist C | January 2019 | January 2020 |
| THSTI, Faridabad | Early Career Medical Research Awardee | July 2016 | January 2019 |
| All India Institute of Medical Sciences, New Delhi | Junior Resident (M.D. Biochemistry) | 2013 | 2016 |
| Private practice | Paediatric consultant | March 2009 | May 2013 |

## 6. Research grants and sponsored projects

### 6.1 Ongoing

| Project | Funding agency | Role | Duration / status | Amount |
|---|---|---|---|---|
| IPD analysis of hypothyroidism in pregnancy: prevalence, risk factors and maternal and fetal outcomes in India (ePMS NHRP-2026-0000226) | Indian Council of Medical Research (ICMR), National Health Research Priority | PI | 20 August 2026 to 19 February 2028 (18 months) | INR 24,51,020 (Year 1 sanctioned) |
| Secondary data analysis of gestational weight gain and maternal-neonatal outcomes in Indian cohorts (ePMS NHRP-2026-0000219) | Indian Council of Medical Research (ICMR), National Health Research Priority | PI | 1 June 2026 to 30 November 2027 (18 months) | INR 19,07,060 (Year 1 sanctioned) |
| Surveillance and Pregnancy/Perinatal Research Platform for India: pregnancy risk stratification data analysis, modelling and capacity building (Research Service Agreement, THSTI P609) | Christian Medical College (CMC) Vellore | PI | 1 October 2025 to 31 March 2028 | INR 1,14,88,500 (excl. GST) |
| Graded Risk-based Assessment for Screening and Prevention of Placental Disorders (GRASPP): randomised controlled trial of a two-step | Indian Council of Medical Research (ICMR) | Co-PI | 11 November 2025 to 10 November 2029 | INR 9,94,20,000 (994.20 lakh) |

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| contingent screening strategy for the great obstetrical syndromes (EMTR/IM-DEL/2025/03-28) | | | | |
| MOMI-India: Multi-Omics for Mothers and Infants, Genomics and Analysis Planning, Phase II (BIRAC/GCI/2023/MOMI-Phase II) | Bill & Melinda Gates Foundation, via Grand Challenges India-BIRAC | Co-PI | 23 August 2023 to 22 August 2027 | INR 10,65,90,000 (1065.90 lakh) |
| Improving maternal and neonatal outcomes using imaging data science (IIRPIG-2023-0001097; File No. EM/Dev/IG/0001097/2023/JJC; PI: Dr Nitya Wadhwa) | ICMR, Extramural Research Programme (Intermediate Grant) | Co-PI (clinical investigation) | 20 March 2024; approved duration 4 years, annual extension after review | INR 2,06,22,808 (Year 1 sanctioned); INR 7,97,46,000 proposed for 4 years |
| Effects of extreme heat on maternal, placental and fetal physiology, lactation and newborn health in India (HiP-India) | Wellcome Trust, UK | Co-PI (clinical domain) | 2024 to 2029 (dates to be confirmed) | GBP 2 million (consortium total) |
| AI for Ultrasound Programme (AI4USG): multicentric evaluation of Indian population-specific tools for antenatal gestational age estimation (MEGA study), with THSTI and IIT Madras | Grand Challenges India (BIRAC - Gates Foundation) | PI (technical domain), validation lead | 2025 to 2026 | Approx. INR 8 crore (incl. supplementary grant) |

### 6.3 Completed

| Project | Funding agency | Role | Duration | Amount |
|---|---|---|---|---|
| Inter-Institutional Programme for Maternal, Neonatal and Infant Sciences: a translational approach to studying preterm birth (GARBH-Ini Phase 2) | DBT, Government of India | Co-Investigator (clinical and epidemiology domain) | 2020 to 2025 | INR 48,85,32,000 (4885.32 lakh) |
| GARBH-Ini Phase 2, THSTI component | DBT, Government of India | Co-PI (clinical and epidemiology domain) | 2020 to 2025 | INR 29,35,56,000 (2935.56 lakh) |
| A bench-to-bedside model for clinical and translational science focused on fetal growth restriction and preterm birth (Glue Grant) | DBT, Government of India | Co-PI (clinical and epidemiology domain) | 2018 to 2023 | INR 6,81,90,000 (681.90 lakh) |
| Multi-Omics for Mothers and Infants (MOMI) Consortium, Phase I | Bill & Melinda Gates Foundation, via Grand Challenges India-BIRAC | Co-PI | 2019 to 2023 | INR 5,16,77,782 (516.78 lakh) |
| COVID-19 Bioresource at the NCR Biotech Science Cluster / DBT Consortium for COVID-19 Research | DBT, Government of India | Co-PI (cohort domain) | 2020 to 2023 | - |
| Preterm birth risk prediction in pregnant women using machine learning models | Grand Challenges India (BIRAC - Gates Foundation) | Co-Investigator | 2020 to 2022 | - |

## 7. Patents (filed in India)

| Title | Application no. | Status | Role |
|---|---|---|---|
| Garbh-Ini Ultrasound Image-based Gestational Age Estimation System (GAUGE) | 202411027837 | Filed 4 April 2024; awarded | Co-conceived the invention, designed and conducted the clinical study, and clinically |

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| | | | evaluated the model. |
| A population-specific gestational dating system and a method thereof | 202341061748 | Filed 13 September 2023 | Conceived the invention, designed the clinical study, clinically phenotyped the participants from whom the model was developed, and clinically evaluated the model. |
| Potently neutralising novel human monoclonal antibodies to SARS-CoV-2 (COVID-19) | 202111038519 | Filed 25 August 2021; awarded | Designed the clinical study and clinically phenotyped the participants from whom the monoclonal antibodies were generated. |
| Immunoassay for detection of Coronaviridae | 202011039696 | Filed 14 September 2020; awarded | Designed the clinical study, clinically phenotyped the participants from whom biospecimens were collected, and clinically evaluated the diagnostic assay. |

## 8. Awards and honours

| Award | Date |
|---|---|
| Award for a PhD student for the Best Published Paper of 2021, THSTI Foundation Day (DBT) | 15 July 2022 |
| Best oral presentation (second prize), Workshop on Air Pollution and Health Effects Research in India, CAPHER-India | 21-23 October 2021 |
| Best PhD oral presentation (second prize), National Science Day, THSTI | 28 February 2021 |
| Dr M.K. Bhan Group Award for the most impactful collaboration, THSTI (as a member of the clinical team of the DBT Consortium for COVID-19 Research) | 2020 |
| Early Career Medical Research Award, THSTI, DBT, Government of India (maternal and child health) | 2017 |
| FIMSA award for best poster, 13th FIMSA Advanced Immunology Course, PGIMER Chandigarh | 17-19 March 2016 |
| Prof. L.M. Shrivastava Medal for best oral presentation, 4th CME of Clinical Biochemistry, Sir Ganga Ram Hospital, New Delhi | 20 March 2015 |
| Certificate of Excellence for best journal club presentation, Department of Biochemistry, AIIMS, New Delhi | 2014 and 2015 |

## 9. Teaching and pedagogy

- Faculty, PhD coursework in epidemiology, clinical research, translational research and biostatistics, BRIC-THSTI (2019-2022 and 2025 onwards).
- Faculty, MSc Clinical Research programme (clinical trials specialisation), THSTI: Epidemiology module (2023 onwards).
- External examiner, PhD thesis evaluation, University of Bergen, Norway (2024).
- External examiner, PhD thesis evaluation, Department of Biochemistry, AIIMS, New Delhi (2026).
- Faculty and co-facilitator, EpiCap, a capacity-building programme for epidemiology teachers in LMICs (ICMR-NIE and University of Bergen) (2023 onwards).

## 10. Institution-building and leadership

- Clinical coordinator and co-organiser, Aryabhata Data Science and AI Program at THSTI (ADAPT), bringing together clinical researchers, data scientists, computational biologists and statisticians.
- Spearheaded the THSTI collaboration with the Robert Bosch Centre for Data Science and AI, IIT Madras: more than INR 10 crore in extramural funding across three projects since 2019, two manuscripts and a patented product (GAUGE).

- Facilitated the Multi-Omics for Mothers and Infants (MOMI) consortium (Gates Foundation via GCI-BIRAC); contributed to the Variable Harmonisation Working Group harmonising clinical data across six cohorts in India, Pakistan, Bangladesh, Zambia and Tanzania (about 25,000 pregnant women).
- Coordinated the clinical research component of the DBT Consortium for COVID-19 Research, including real-world effectiveness evaluation of Covishield and Covaxin against Delta and Omicron that informed the national immunisation programme.
- Lead, THSTI pregnancy risk stratification data-analysis and capacity-building services to CMC Vellore (2025-2028).
- Coordinator, day-to-day activities of THSTI AI and imaging research projects in the Maternal and Child Health Programme (ICMR imaging data science project and Grand Challenges India AI for Ultrasound Programme), 2026 onwards.

## 11. Memberships, editorial and expert roles

- Member, Scientific Advisory Committee, ICMR-National Institute of Nutrition, Hyderabad.
- Expert member, Project Selection Committee (Epidemiology), ICMR intramural funding, 2024 onwards.
- Academic Editor, PLOS Global Public Health, and Associate Editor, Frontiers in Pediatrics, 2023 onwards.
- Member, Technical Advisory Group, Obesity Individual Participant Data Meta-analysis, ICMR (Division of Reproductive, Child Health and Nutrition), 2026 onwards.
- Member, Mentorship Advisory Committee, PRERNA 3.0, Society for Applied Studies, New Delhi, 2026 onwards.
- Expert reviewer, ICMR-RCN Division intramural research proposals.

## 12. Workshops organised

- Data science-driven solutions to improve maternal and child health: improving accuracy of pregnancy dating using AI, IIT Madras, Chennai, 22 February 2023 (co-organiser).
- Stakeholder meeting on AI-enabled pregnancy dating models and ultrasound tools, THSTI, April 2023 (co-organiser).
- Annual workshops on Artificial Intelligence in Healthcare with the University of Oxford: 5-6 March 2020 (Delhi), 21-24 November 2021 (online), 12-14 June 2022 (Goa) (co-organiser).
- Faculty Development Programme on clinical research methodology, Sharda University, Greater Noida, 8 February 2020 (co-organiser).

## 13. Book chapters and technical documents

- Essential Diagnostics List 2019-20 for the World Health Organization, South-East Asia Region (co-author).

## 14. Bibliometric summary

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| **Articles in PubMed (past 10 years)** | 56 (PubMed, as on 28 August 2026) |
| **h-index** | 22 (Google Scholar, as on 28 August 2026) |
| **Citations** | 3,841 (Google Scholar, as on 28 August 2026); i10-index 31 |
| **Patents** | 4 Indian patent applications (3 awarded, 1 filed) |
| **Fellowships of academies** | Nil |

## 15. Publications (60, peer-reviewed, reverse chronological, AMA style)

*Papers with more than six authors list the first three authors followed by et al. The bracket gives the role of Dr Thiruvengadam.*

1. Chanian R, Thiruvengadam R, Desiraju BK. Ensuring safe and responsible use of artificial intelligence in medical imaging: pitfalls and practical solutions through the lens of obstetric ultrasound. *Indian J Radiol Imaging*. 2026;36(4):614-620. doi:10.1055/s-0046-1829015. PMID: 42921404 *[Second author]*

2. Gopal P, Opondo C, Mukherjee D, et al; ICMR-SPIC Consortium. Risk of stillbirth among pregnant women with anaemia in India: secondary analysis of multiple cohorts from the ICMR-SPIC consortium. *Lancet Reg Health Southeast Asia*. 2026;52:100825. doi:10.1016/j.lansea.2026.100825. PMID: 42572652 *[Co-author]*

3. Iqbal N, Murugan M, Subramanyam S, et al. Artificial intelligence-augmented standardized patient models for AETCOM (Attitude, Ethics, and Communication) competency evaluation: a pilot study. *Cureus*. 2026;18(7):e112683. doi:10.7759/cureus.112683. PMID: 42602116 *[Last author]*

4. Mary Kurian C, Kumar K, Thomas D, et al; HiP-India Group. To gain insights into heat adaptations among women during the antenatal period in India - the Heat in Pregnancy Project (HiP-I) India - qualitative study protocol. *Wellcome Open Res*. 2026;11:93. doi:10.12688/wellcomeopenres.25507.2. PMID: 42064355 *[Co-author]*

5. Tang X, Ahmed S, Ahmed FB, et al; MOMI Consortium. Multi-Omics for Mothers and Infants (MOMI) Consortium: a global initiative to study adverse pregnancy outcomes. *J Glob Health*. 2026;16:05002. doi:10.7189/jogh.16.05002. PMID: 41926172 *[Group/consortium member]*

6. Kotamreddy S, Srinivasan R, Sharma N, et al. Determinants of small vulnerable newborn births in two rural community-based cohorts in India: a prospective study. *Lancet Reg Health Southeast Asia*. 2026;47:100739. doi:10.1016/j.lansea.2026.100739. PMID: 41799251 *[Co-author]*

7. Chanian R, Thiruvengadam R, Desiraju BK. Prediction of spontaneous preterm birth in the second trimester: current models, limitations, and the way forward. *Indian J Pediatr*. 2026;93(8):852-858. doi:10.1007/s12098-026-06030-1. PMID: 41729215 *[Second author]*

8. Cavoretto PI, Villar J, Farina A, et al; INTERCOVID Consortium. COVID-19 vaccination status during pregnancy and preeclampsia risk: the pandemic-era cohort of the INTERCOVID consortium. *EClinicalMedicine*. 2026;93:103785. doi:10.1016/j.eclinm.2026.103785. PMID: 41938842 *[Group/consortium member]*

9. Manna S, Roy SP, Lyngdoh T, et al. Identifying research priorities for gestational weight gain in India: a national exercise. *Indian J Med Res*. 2026;163(2):215-219. doi:10.25259/IJMR_2976_2025. PMID: 41949129 *[Co-author]*

10. Jakhar K, Sonar S, Singh G, et al. SARS-CoV-2 spike antibodies cross-react with dengue virus and enhance infection. *Front Immunol*. 2026;16:1724625. doi:10.3389/fimmu.2025.1724625. PMID: 41624894 *[Co-author]*

11. Thiruvengadam R, Karthikeyan G. Interpreting cardiovascular disease burden for low-middle income countries: a perspective from India. *J Am Coll Cardiol*. 2025;86(22):2112-2114. doi:10.1016/j.jacc.2025.10.014. PMID: 41298020 *[First author]*

12. Thiruvengadam R, Raghavan S, Shanmugam R, et al; HiP-India Group. Cohort profile for the Heat in Pregnancy-India (HiP-India) Study. *Wellcome Open Res*. 2025;10:472. doi:10.12688/wellcomeopenres.24393.2. PMID: 41623503 *[First author]*

13. Bhattacharjee E, Thiruvengadam R, Varshney D, et al; GARBH-Ini Study Group. Deciphering mechanisms leading to preterm birth through post-GWAS integration of multi-omics data. *iScience*. 2025;28(11):113861. doi:10.1016/j.isci.2025.113861. PMID: 41362759 *[Second author]*

14. Raghavan S, Isaac TS, Arya D, et al; HiP-India Group. Association between heat and air pollution (PM2.5 and black carbon) exposure in pregnancy and preterm birth in low- and middle-income countries: a systematic review and meta-analysis. *BMJ Paediatr Open*. 2025;9(1):e003428. doi:10.1136/bmjpo-2025-003428. PMID: 40953917 *[Co-author]*

15. ICMR-SPIC Consortium; Mukherjee R. Cohort profile of the ICMR-Stillbirth Pooled India Cohort (ICMR-SPIC): estimating prevalence, analysing risk factors and developing prediction models for stillbirths in India. *BMJ Open*. 2025;15(8):e099401. doi:10.1136/bmjopen-2025-099401. PMID: 40854841 *[Group/consortium member]*

16. Dinakaran A, Thiruvengadam R, Srinivasan AR, et al. Prediction models for gestational diabetes mellitus: diagnostic utility of clinical and biochemical markers. *Indian J Endocrinol Metab*. 2025;29(3):295-302. doi:10.4103/ijem.ijem_497_24. PMID: 40688618 *[Second author]*

17. Talukdar D, Raju YJ, Jana P, et al; GARBH-Ini Study Group. Genomic insights into the potency and functional roles of Lactobacillus species in term and preterm births. *Genomics*. 2025;117(4):111063. doi:10.1016/j.ygeno.2025.111063. PMID: 40456421 *[Group/consortium member]*

18. Sebastião YV, Thiruvengadam R, Khanam R, et al. Association between gestational weight gain and adverse pregnancy outcomes: cohort analysis from South Asia and sub-Saharan Africa. *BMJ Public Health*. 2025;3(1):e000900. doi:10.1136/bmjph-2024-000900. PMID: 40433067 *[Second author]*

19. Kharbanda N, Biswas A, Tiwari A, et al. Placental proteomics reveals an elevated level of aldo-keto reductase 1-B1, highlighting its potential role in spontaneous preterm birth. *J Proteome Res*. 2025;24(2):612-623. doi:10.1021/acs.jproteome.4c00698. PMID: 39762117 *[Co-author]*

20. Talukdar D, Sarkar M, Ahrodia T, et al. Previse preterm birth in early pregnancy through vaginal microbiome signatures using metagenomics and dipstick assays. *iScience*. 2024;27(11):111238. doi:10.1016/j.isci.2024.111238. PMID: 39569373 *[Co-author]*

21. Thiruvengadam R, Ayushi, Murugesan DR, et al; GARBH-Ini Study Team. Incidence of and risk factors for small vulnerable newborns in north India: a secondary analysis of a prospective pregnancy cohort. *Lancet Glob Health*. 2024;12(8):e1261-e1277. doi:10.1016/S2214-109X(24)00212-2. PMID: 39030058 *[First author]*

22. Gadekar VP, Damaraju N, Xavier A, et al; GARBH-Ini Study Group. Development and external validation of Indian population-specific Garbhini-GA2 model for estimating gestational age in second and third trimesters. *Lancet Reg Health Southeast Asia*. 2024;25:100362. doi:10.1016/j.lansea.2024.100362. PMID: 39021476 *[Corresponding author]*

23. Ramasamy T, Varughese B, Singh M, et al. Post-natal gestational age assessment using targeted metabolites of neonatal heel prick and umbilical cord blood: a GARBH-Ini cohort study from North India. *J Glob Health*. 2024;14:04115. doi:10.7189/jogh.14.04115. PMID: 38968007 *[Co-author]*

24. Gleed AD, Mishra D, Self A, et al. Statistical characterisation of fetal anatomy in simple obstetric ultrasound video sweeps. *Ultrasound Med Biol*. 2024;50(7):985-993. doi:10.1016/j.ultrasmedbio.2024.03.006. PMID: 38692940 *[Co-author]*

25. Wadhwa K, Malik S, Balaji S, et al. Correlation of severity and clinical outcomes of COVID-19 with virus variants: a prospective, multicentre hospital network study. *Indian J Med Res*. 2024;159(1):91-101. doi:10.4103/ijmr.ijmr_1041_23. PMID: 38344919 *[Co-author]*

26. Binayke A, Zaheer A, Vishwakarma S, et al. A quest for universal anti-SARS-CoV-2 T cell assay: systematic review, meta-analysis, and experimental validation. *NPJ Vaccines*. 2024;9(1):3. doi:10.1038/s41541-023-00794-9. PMID: 38167915 *[Co-author]*

27. Monangi NK, Xu H, Fan YM, et al; INTERBIO-21st Study Consortium. Association of maternal prenatal copper concentration with gestational duration and preterm birth: a multicountry meta-analysis. *Am J Clin Nutr*. 2024;119(1):221-231. doi:10.1016/j.ajcnut.2023.10.011. PMID: 37890672 *[Co-author]*

28. Thiruvengadam R, Rizvi ZA, Raghavan S, et al. Clinical and experimental evidence suggest omicron variant of SARS-CoV-2 is inherently less pathogenic than delta variant independent of previous immunity. *Eur J Med Res*. 2023;28(1):421. doi:10.1186/s40001-023-01373-3. PMID: 37821945 *[First author]*

29. Bhattacharjee E, Thiruvengadam R, Ayushi, et al; GARBH-Ini Team. Genetic variants associated with spontaneous preterm birth in women from India: a prospective cohort study. *Lancet Reg Health Southeast Asia*. 2023;14:100190. doi:10.1016/j.lansea.2023.100190. PMID: 37492417 *[Second author]*

30. Das J, Wadhwa N, Natchu UC, et al. Genome-wide temporal landscaping of DNA methylation in pregnant women delivering at term: a GARBH-InI study. *Epigenomics*. 2023;15(9):543-556. doi:10.2217/epi-2023-0145. PMID: 37345372 *[Co-author]*

31. Batra G, Murugesan DR, Raghavan S, et al. Trends of humoral immune responses to heterologous antigenic exposure due to vaccination and omicron SARS-CoV-2 infection: implications for boosting. *Indian J Med Res*. 2023;157(6):509-518. doi:10.4103/ijmr.ijmr_2521_22. PMID: 37322634 *[Co-author]*

32. Thiruvengadam R, Desiraju BK, Sachdev HS, Bhatnagar S. The challenges in gestational weight gain monitoring in low and middle income settings. *Eur J Clin Nutr*. 2023;77(7):764-765. doi:10.1038/s41430-023-01292-w. PMID: 37316558 *[First author]*

33. Gupta V, Singh A, Ganju S, et al. Severity and mortality associated with COVID-19 among children hospitalised in tertiary care centres in India: a cohort study. *Lancet Reg Health Southeast Asia*. 2023;13:100203. doi:10.1016/j.lansea.2023.100203. PMID: 37159588 *[Co-author]*

34. Deshpande S, Ansari MY, Sutar J, et al. Ancestral SARS-CoV-2-driven antibody repertoire diversity in an unvaccinated individual correlates with expanded neutralization breadth. *Microbiol Spectr*. 2023;11(2):e0433222. doi:10.1128/spectrum.04332-22. PMID: 36946746 *[Co-author]*

35. Self A, Chen Q, Desiraju BK, et al; CALOPUS Study Group. Developing clinical artificial intelligence for obstetric ultrasound to improve access in underserved regions: protocol for a Computer-Assisted Low-Cost Point-of-Care UltraSound (CALOPUS) study. *JMIR Res Protoc*. 2022;11(9):e37374. doi:10.2196/37374. PMID: 36048518 *[Co-author]*

36. Munblit D, Nicholson T, Akrami A, et al; PC-COS project steering committee. A core outcome set for post-COVID-19 condition in adults for use in clinical practice and research: an international Delphi consensus study. *Lancet Respir Med*. 2022;10(7):715-724. doi:10.1016/S2213-2600(22)00169-2. PMID: 35714658 *[Group/consortium member]*

37. Vikkurthi R, Ansari A, Pai AR, et al. Inactivated whole-virion vaccine BBV152/Covaxin elicits robust cellular immune memory to SARS-CoV-2 and variants of concern. *Nat Microbiol*. 2022;7(7):974-985. doi:10.1038/s41564-022-01161-5. PMID: 35681012 *[Co-author]*

38. Hingankar N, Deshpande S, Das P, et al. A combination of potently neutralizing monoclonal antibodies isolated from an Indian convalescent donor protects against the SARS-CoV-2 Delta variant. *PLoS Pathog*. 2022;18(4):e1010465. doi:10.1371/journal.ppat.1010465. PMID: 35482816 *[Co-author]*

39. Giuliani F, Oros D, Gunier RB, et al. Effects of prenatal exposure to maternal COVID-19 and perinatal care on neonatal outcome: results from the INTERCOVID Multinational Cohort Study. *Am J Obstet Gynecol*. 2022;227(3):488.e1-488.e17. doi:10.1016/j.ajog.2022.04.019. PMID: 35452653 *[Co-author]*

40. Garg PK, Thiruvengadam R. Effectiveness of SARS-CoV-2 vaccines in the post-natural infection world. *Lancet Infect Dis*. 2022;22(6):745-747. doi:10.1016/S1473-3099(22)00207-9. PMID: 35366961 *[Second author]*

41. Thiruvengadam R, Awasthi A, Bhatnagar S, Garg PK; DBT Consortium for COVID-19 Research. Effectiveness of ChAdOx1 nCoV-19 vaccine during the delta (B.1.617.2) variant surge in India: authors' reply. *Lancet Infect Dis*. 2022;22(4):447. doi:10.1016/S1473-3099(22)00120-7. PMID: 35338864 *[First author]*

42. Kaushal K, Ayushi, Sharma N, et al; GARBH-Ini Study Group. Longitudinal changes in serum immune markers during normal pregnancy in a North-Indian population. *Am J Reprod Immunol*. 2022;87(5):e13531. doi:10.1111/aji.13531. PMID: 35312132 *[Corresponding author]*

43. Medigeshi GR, Batra G, Murugesan DR, et al. Sub-optimal neutralisation of omicron (B.1.1.529) variant by antibodies induced by vaccine alone or SARS-CoV-2 infection plus vaccine (hybrid immunity) post 6-months. *EBioMedicine*. 2022;78:103938. doi:10.1016/j.ebiom.2022.103938. PMID: 35305396 *[Co-author]*

44. Munblit D, Nicholson TR, Needham DM, et al. Studying the post-COVID-19 condition: research challenges, strategies, and importance of Core Outcome Set development. *BMC Med*. 2022;20(1):50. doi:10.1186/s12916-021-02222-y. PMID: 35114994 *[Co-author]*

45. Thiruvengadam R, Awasthi A, Medigeshi G, et al; DBT India Consortium for COVID-19 Research. Effectiveness of ChAdOx1 nCoV-19 vaccine against SARS-CoV-2 infection during the delta (B.1.617.2) variant surge in India: a test-negative, case-control study and a mechanistic study of post-vaccination immune responses. *Lancet Infect Dis*. 2022;22(4):473-482. doi:10.1016/S1473-3099(21)00680-0. PMID: 34838183 *[First author]*

46. Thiruvengadam R, Binayke A, Awasthi A. SARS-CoV-2 delta variant: a persistent threat to the effectiveness of vaccines. *Lancet Infect Dis*. 2022;22(3):301-302. doi:10.1016/S1473-3099(21)00697-6. PMID: 34826382 *[First author]*

47. Thiruvengadam R, Desiraju BK, Natchu UCM, et al; GARBH-Ini Study Team. Gestational weight gain trajectories in GARBH-Ini pregnancy cohort in North India and a comparative analysis with global references. *Eur J Clin Nutr*. 2022;76(6):855-862. doi:10.1038/s41430-021-01040-y. PMID: 34785811 *[First author]*

48. Murugavelu P, Perween R, Shrivastava T, et al. Non-neutralizing SARS CoV-2 directed polyclonal antibodies demonstrate cross-reactivity with the HA glycans of influenza virus. *Int Immunopharmacol*. 2021;99:108020. doi:10.1016/j.intimp.2021.108020. PMID: 34426117 *[Co-author]*

49. Papageorghiou AT, Deruelle P, Gunier RB, et al. Preeclampsia and COVID-19: results from the INTERCOVID prospective longitudinal study. *Am J Obstet Gynecol*. 2021;225(3):289.e1-289.e17. doi:10.1016/j.ajog.2021.05.014. PMID: 34187688 *[Co-author]*

50. Kumar S, Kumari N, Talukdar D, et al; GARBH-Ini Study Group. The vaginal microbial signatures of preterm birth delivery in Indian women. *Front Cell Infect Microbiol*. 2021;11:622474. doi:10.3389/fcimb.2021.622474. PMID: 34094994 *[Co-author]*

51. Thiruvengadam R, Chattopadhyay S, Mehdi F, et al; DBT India Consortium for COVID-19 Research. Longitudinal serology of SARS-CoV-2-infected individuals in India: a prospective cohort study. *Am J Trop Med Hyg*. 2021;105(1):66-72. doi:10.4269/ajtmh.21-0164. PMID: 34003792 *[First author]*

52. Vijayram R, Damaraju N, Xavier A, et al; GARBH-Ini Study Group. Comparison of first trimester dating methods for gestational age estimation and their implication on preterm birth classification in a North Indian cohort. *BMC Pregnancy Childbirth*. 2021;21(1):343. doi:10.1186/s12884-021-03807-4. PMID: 33931016 *[Co-author]*

53. Villar J, Ariff S, Gunier RB, et al. Maternal and neonatal morbidity and mortality among pregnant women with and without COVID-19 infection: the INTERCOVID Multinational Cohort Study. *JAMA Pediatr*. 2021;175(8):817-826. doi:10.1001/jamapediatrics.2021.1050. PMID: 33885740 *[Co-author]*

54. Mehdi F, Chattopadhyay S, Thiruvengadam R, et al. Development of a fast SARS-CoV-2 IgG ELISA, based on receptor-binding domain, and its comparative evaluation using temporally segregated samples from RT-PCR positive individuals. *Front Microbiol*. 2021;11:618097. doi:10.3389/fmicb.2020.618097. PMID: 33552028 *[Co-author]*

55. Kumar B, Dey AK, Saha S, et al; GARBH-Ini Study Group. Dynamic alteration in the vaginal secretory proteome across the early and mid-trimesters of pregnancy. *J Proteome Res*. 2021;20(2):1190-1205. doi:10.1021/acs.jproteome.0c00433. PMID: 33497241 *[Co-author]*

56. Chaudhuri S, Thiruvengadam R, Chattopadhyay S, et al; DBT India Consortium for COVID-19 Research. Comparative evaluation of SARS-CoV-2 IgG assays in India. *J Clin Virol*. 2020;131:104609. doi:10.1016/j.jcv.2020.104609 *[Second author]*

57. Dey AK, Kumar B, Singh AK, et al; GARBH-Ini Study Group. Salivary proteome signatures in the early and middle stages of human pregnancy with term birth outcome. *Sci Rep*. 2020;10(1):8022. doi:10.1038/s41598-020-64483-6 *[Co-author]*

58. Mehta O, Ghosh TS, Kothidar A, et al; GARBH-Ini Study Group. Vaginal microbiome of pregnant Indian women: insights into the genome of dominant Lactobacillus species. *Microb Ecol*. 2020;80(2):487-499. doi:10.1007/s00248-020-01501-0 *[Co-author]*

59. Kumar S, Kumar R, Khan L, et al. CD4-binding site directed cross-neutralizing scFv monoclonals from HIV-1 subtype C infected Indian children. *Front Immunol*. 2017;8:1568. doi:10.3389/fimmu.2017.01568 *[Co-author]*

60. Khan L, Kumar R, Thiruvengadam R, et al. Cross-neutralizing anti-HIV-1 human single chain variable fragments (scFvs) against CD4 binding site and N332 glycan identified from a recombinant phage library. *Sci Rep*. 2017;7:45163. doi:10.1038/srep45163 *[Co-author]*

*Data as of 5 November 2026. Version: 2026-11-05 (master CV v2026.10 + ICMR sanction letter for IIRPIG-2023-0001097) (public edition)*